

Single Core Copper Conductor, XLPE insulated

Lead Sheathed and HDPE Sheath  
(CU/XLPE/LEAD/HDPE)

Voltage Grade

127 / 220 (245) KV

Description

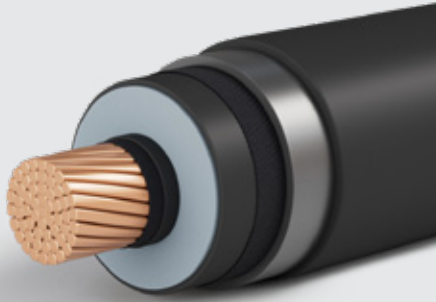
Stranded circular or segmental compacted copper conductor Filled with swelling powder between conductor layers, Semi-conductive water blocking tape over conductor to protect the Conductor from longitudinal water penetration, semi-conducting layer as conductor screen, XLPE insulated, semi-conducting layer as non metallic insulation screen, Semi-conductive water blocking tape to protect the screen area from longitudinal water penetration, extruded lead sheathed with a suitable thickness to withstand the required earth fault current, and also act as a radial water pentration protection for the cable and HDPE sheathed with graphite coating or extruded semi-conducting layer for the D.C Sheath test.

Application

These cables are generally suitable for direct burial or for installation on trays or in ducts.

Technical Data

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Relevant Standard       | EC 60228, 62067, IEC 60229 and 60811                                                                                                                                                                                                                                                                                                                                                                                                              |
| Conductor               | <p>Plain copper conductor class 2</p> <p>Circular Comnpacted (applied for C.S.A up to 800 mm2) or Segmental Compacted (Applied from C.S.A = 1000 mm2 and above)</p> <p>Five segments of compacted conductor in sector shape of 72° are assembled together with separation of non-metallic tapes to reduce the skin effect which reduce the Rac.</p>                                                                                               |
| Conductor Screen        | Extruded Semiconductive Compound                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Insulation              | Extruded XLPE Compound Rated 90° C                                                                                                                                                                                                                                                                                                                                                                                                                |
| Insulation Screen       | <p><b>Non Metallic part:</b> Extruded Semiconductive Compound (Bonded) to the insulation</p> <p><b>Metallic part:</b> Extruded Lead Sheath with a suitable thickness</p> <p>Conductor screen, XLPE insulation and non metallic insulation screen are applied at the same time using triple head extruder.</p>                                                                                                                                     |
| Outer Sheath            | <p>An extruded layer of HDPE shall be applied over the metallic layers for protection against chemical reaction &amp; mechanical protection.</p> <p>Extruded semi conductive layer or graphite coating shall be applied over the outer sheath for D.C testing purpose</p>                                                                                                                                                                         |
| Water Pentration layers | <p>i) <b>Longitudinal Water Penetration:</b></p> <p><b>1) the conductor:</b> the conductor layers filled with swelling powder and overlaped semiconductive water blocking tape over the conductor</p> <p><b>2) The Metallic part of the insulation screen:</b> An overlaped semiconductive water blocking tape before the metallic screen</p> <p>ii) <b>Radial Water Penetration:</b> Extruded lead sheath applied before the PE outer Sheath</p> |
| Outer Sheath Color      | Black (default color), or any other color according to customer request                                                                                                                                                                                                                                                                                                                                                                           |

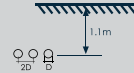


Maximum Operating Temp. 90° C

Minimum bending Radius (16-18) x outer diameter

Packing Condition Our standard length on Steel Drum / or according to customer request

| Product Code                                       | Nominal Cross Sectional Area | Conductor   |                                                                                     |                                                                                     | Operating Capacitance | Inductance     |         | Thickness of Conductor Screen | Thickness of Insulation | Thickness of Conductor Screen | Thickness of Insulation | Approx. Overall Diameter | Approx. Weight |
|----------------------------------------------------|------------------------------|-------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------|----------------|---------|-------------------------------|-------------------------|-------------------------------|-------------------------|--------------------------|----------------|
|                                                    |                              | DC at 20 °C | AC at 90 °C                                                                         |                                                                                     |                       | Flat seperated | Trefoil |                               |                         |                               |                         |                          |                |
|                                                    |                              |             |  |  |                       |                |         |                               |                         |                               |                         |                          |                |
|                                                    |                              |             |                                                                                     |                                                                                     |                       |                |         |                               |                         |                               |                         |                          |                |
|                                                    |                              | mm²         | Ω/km                                                                                | Ω/km                                                                                |                       | μF/km          | mH/km   |                               |                         |                               |                         |                          |                |
| Electrical & Construnctional data (Nominal Values) |                              |             |                                                                                     |                                                                                     |                       |                |         |                               |                         |                               |                         |                          |                |
| PE-CM26-XF-01-AH                                   | 1000 S                       | 0.0176      | 0.0246                                                                              | 0.0248                                                                              | 0.1888                | 0.5863         | 0.4015  | 1.5                           | 25.0                    | 1.5                           | 4.5                     | 118.0                    | 28560          |
| PE-CM27-XF-01-AH                                   | 1200 S                       | 0.0151      | 0.0218                                                                              | 0.0220                                                                              | 0.1963                | 0.5784         | 0.3935  | 1.5                           | 25.0                    | 1.5                           | 4.5                     | 120.6                    | 30370          |

| Product Code                                                                           | Nominal Cross Sectional Area | Laying conditions: Flat formationa |                                                                                     |                                                                                     |
|----------------------------------------------------------------------------------------|------------------------------|------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|                                                                                        |                              | Type of Earthing Bonding System    | Direct burial                                                                       | in Air (Shaded)                                                                     |
|                                                                                        |                              |                                    |  |  |
|                                                                                        |                              |                                    | $\rho_t=120$<br>$T = 35\text{ }^{\circ}\text{C}$                                    | $T = 40\text{ }^{\circ}\text{C}$                                                    |
| Continuous current ratings (load factor = 100%) for one circuit in operation (Amperes) |                              |                                    |                                                                                     |                                                                                     |
| PE-CM26-XF-01-AH                                                                       | 1000 S                       | Single End Bonded or Cross Bonding | 1005                                                                                | 1531                                                                                |
| PE-CM27-XF-01-AH                                                                       | 1200 S                       |                                    | 1075                                                                                | 1655                                                                                |

The above data is approximate and subject to manufacturing tolerances

Note:-

Flame retardant PVC sheathed or MDPE, LLDPE Sheathed are available according to customer request

The Metallic Screen is designed to carry an earth fault cureent = 63 KA per one seond per three pahses, any other design is also available as per customer request